Joseph McGarry

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SKILLS

JavaScript, TypeScript, Python, React, Next.js, React Native, Expo, Node.js, Express.js, Redux, Redux Toolkit, GraphQL, Bun, MongoDB, Mongoose, Redis, PostgreSQL, SQL, SQLite, FAISS, AWS, Docker, Vercel CI/CD, GitHub Actions, Git, REST API, OpenAI API, OpenAIEmbeddings, LangChain, Prompt engineering, RAG (Retrieval-Augmented Generation), Large Language Models (LLMs), AJAX, HTML, CSS, SCSS, Tailwind CSS, Webpack, Vite, Jest, Supertest, React Testing Library

EXPERIENCE

AI Developer

DataAnnotation **June 2024 - Present, Remote**

- Trained large language models to generate deployment-ready JavaScript, TypeScript, Python, and HTML/CSS code by crafting precise prompt templates and iteratively fine-tuning outputs, resulting in a 30% reduction in error rates.
- Reviewed and corrected AI-generated content against client-specific safety and truthfulness criteria by executing targeted validation protocols, cutting harmful or irrelevant outputs by 40% and boosting overall reliability.
- Prompted models with complex coding scenarios and debugged 20% of generated solutions, which honed functional robustness and drove a 30% improvement in model performance across use cases.
- Documented insights across a custom 7-axis evaluation framework to guide iterative enhancements, enabling a 20% reduction in output inconsistencies and aligning stakeholder expectations.
- Identified and resolved multi-language code defects by applying cross-disciplinary problem-solving techniques, ensuring 100% compatibility across varied programming tasks and elevating system stability.
- Facilitated multi-turn AI interactions to fine-tune contextual responses in code, text, and image modalities, delivering an uplift in client satisfaction.

Full Stack Software Engineer

OS-Labs **December 2023 - June 2024, Remote**

- Leveraged Bun's ultra-fast runtime optimizations to build our GraphQL caching solution, slashing build times by 400% and accelerating developer velocity.
- Integrated a Redis caching layer with an LFU invalidation policy into our middleware, reducing database load and improving p95 response times by 1400 %.
- Developed a React-based UI for query management and visualizations, driving an increase in adoption and streamlining developer workflows.
- Engineered backend services using Express.js and MongoDB to parse, cache, and scale GraphQL queries, consistently maintaining sub-100 ms response times under load.
- Deployed a fully automated CI/CD pipeline on Vercel for project documentation, eliminating 100 % of manual deployment tasks and ensuring instant updates.
- Led weekly agile ceremonies with an international team of four engineers, achieving an 85 % sprint completion rate and fostering cross-functional collaboration.

PROJECT

Employee Handbook QA Agent

github.com/Joseph-McGarry/AgentDemo

- Built a retrieval-augmented QA pipeline over a PDF with LangChain and FAISS, returning grounded answers with page-level citations and a strict refusal policy for out-of-scope queries.
- Increased recall and intent match by normalizing queries using a regex and LLM alias resolver, cached in a JSON file and auto-refreshed when the source PDF changes.
- Improved answer quality with an MMR-based retriever and contextual compression to surface diverse, high-signal passages while trimming noise.
- Added an optional hybrid retriever (BM25 plus vector ensemble) for acronym-heavy sections; tuned top-k and the prompt to emphasize quotes and citations.
- Implemented a two-stage alias extractor (regex "Full (ACR)" patterns plus LLM enrichment) with normalization heuristics (title case, filler removal, length caps) to produce clean mappings from acronyms to their expanded forms.
- Hardened developer experience with .env secrets, centralized settings (pathlib), modular utilities (embeddings, prompts, alias_extractor, query_helpers) and a CLI REPL for rapid evaluation.

Tessellator | Tessellation Design Tool

josephmcgarry.dev/projects/tessellations

- Architected dynamic grid state in React with hooks and refs, enabling real-time updates across a 20×20 cell canvas with <50 ms render latency.
- Engineered offscreen canvas composition and per-cell rotation transforms to sustain 60 FPS tessellation rendering under heavy draw loads.
- Integrated html2canvas for seamless, high-resolution PNG exports directly from the UI without blocking the main render thread.
- Consolidated disparate mouse/touch handlers into a unified event pipeline, reducing handler code by 70% and ensuring consistent drag-and-paint interactions.
- Composed a Tailwind CSS control suite (color picker, sliders, toggles) to drive grid parameters, decoupling UI layout from business logic for easier iteration.

EDUCATION

Bachelor of Arts

Brevard College • NC

Advanced Engineering Residency

Codesmith • 2023–2024